

Ex:no:12

Experiment to detect outliers in a given data set.

AIM:

To identify and remove outliers from a given dataset using the **IQR (Interquartile Range) method**.

PROGRAM DESCRIPTION:

The program generates a random dataset and calculates the **25th, 50th and 75th percentiles**. The lower range (LR) and upper range (UR) are calculated using IQR. Values outside these ranges are identified as outliers and removed from the dataset.

PROGRAM:

```
import numpy as np

import matplotlib.pyplot as plt

array=np.random.randint(10,120,18)

array

array.mean()

np.percentile(array,25)

np.percentile(array,50)

np.percentile(array,75)

np.percentile(array,100)

#outliers detection

def outDetection(array):

    sorted(array)

    Q1,Q3=np.percentile(array,[25,75])

    IQR=Q3-Q1

    lr=Q1-(1.5*IQR)

    ur=Q3+(1.5*IQR)

    return lr,ur

lr,ur=outDetection(array)

lr,ur
```

```

import seaborn as sns

%matplotlib inline

sns.displot(array)

plt.show()

sns.distplot(array)

plt.show()

new_array=array[(array>lr) & (array<ur)]

new_array

sns.displot(new_array)

plt.show()

lr1,ur1=outDetection(new_array)

lr1,ur1

final_array=new_array[(new_array>lr1) & (new_array<ur1)]

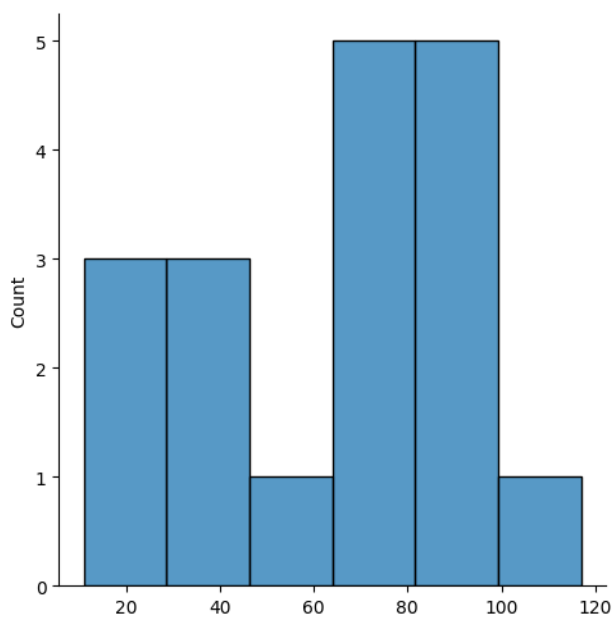
final_array

sns.distplot(final_array)

plt.show()

```

OUTPUT:



/tmp/ipykernel_608/3911722544.py:31: UserWarning:

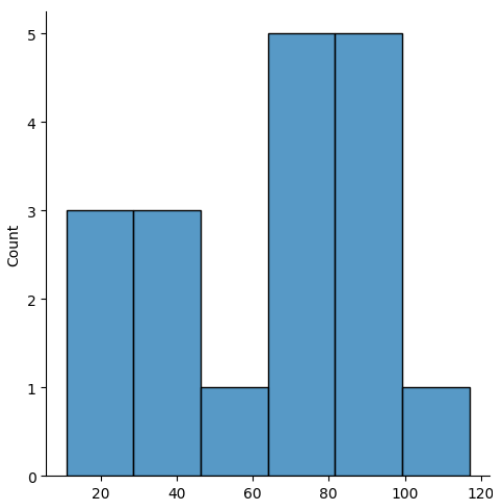
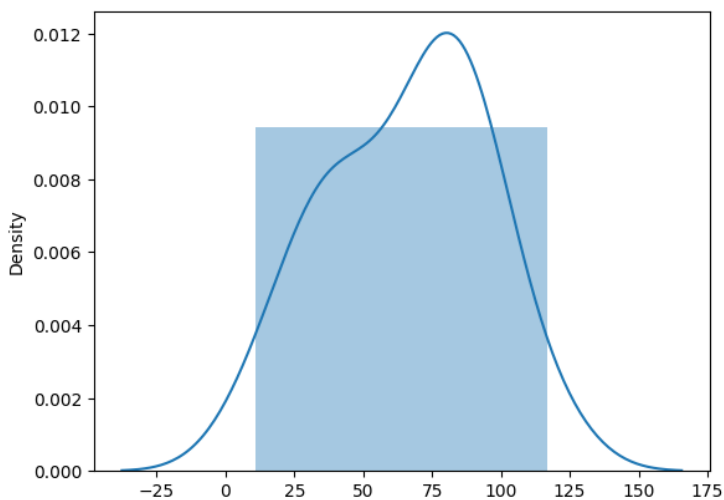
`sns.distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `sns.displot` (a figure-level function with similar flexibility) or `sns.histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see

<https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(array)
```



/tmp/ipykernel_608/3911722544.py:46: UserWarning:

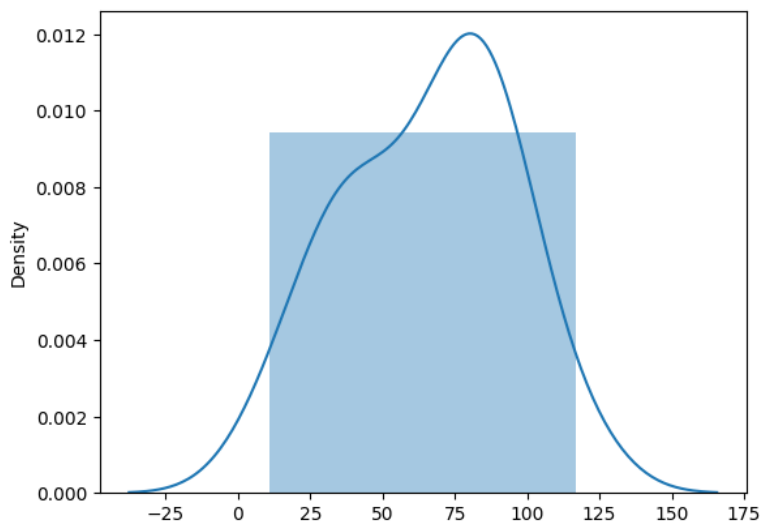
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```
sns.distplot(final_array)
```



RESULT:

Thus, the outliers in the given dataset were successfully detected and removed using the **IQR method**, and the final dataset was obtained.